→ Basics of electrical Engineering

Unit → 1

– Resistance

Opposition offered by a material to flow of electric current

$R = \rho \frac{l}{A}$ By ohm's law, $R = \frac{V}{I}$

→ Ohm's Law

Acc to ohm's law, potential diff across any 2 points of a conductor will be directly proportional to the current flowing through it when physical conditio

$V \propto I$ like temp

$V = IR$

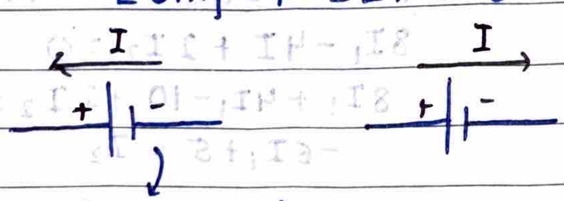
→ Kirchoff's Laws

• KVL (Based on conservation of energ

In any closed ckt (mesh) the alge

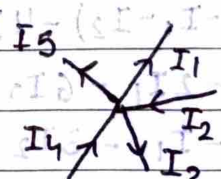
emf & voltage drops is zero.

$\Sigma Emf + \Sigma IR = 0$



→ Drop in potential (-ve)

Rise in potential (+ve)



$\sum_{i=1}^{5} I_i = 0$

• KCL (Conservation of charges)

The algebraic sum of all currents meeting at a pt or junction will be zero

$\Sigma I = 0$

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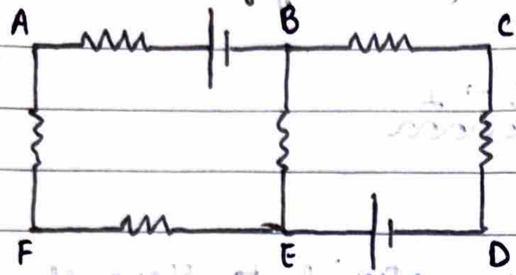
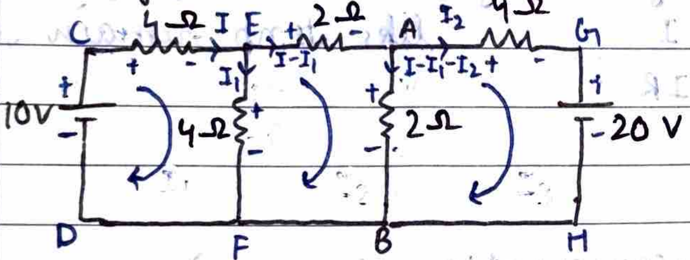
→ ## Mesh Analysis method



Mesh is a closed path on electrical circuit via which current can flow but there is no other closed path in it like ABEF & BCDE → Mesh
But loop can contain closed path within it too like ABEF, BCDE & ACDF → Loop

Q. Calculate current in AB of 2 Ω using mesh analysis



In mesh CEDF
Applying KVL

$$10 - 4I - 4I_1 = 0$$
$$4I + 4I_1 = 10$$
$$4I = 10 - 4I_1$$

In AGHB

$$2(I - I_1 - I_2) - 4I_2 - 20 = 0$$
$$2I - 2I_1 - 6I_2 = 20$$
$$I - I_1 - 3I_2 = 10$$
$$\frac{5-2I_1}{2} - I_1 - 15 + 18I_1 = 10$$
$$5 - 2I_1 + 36I_1 + 30 = 20$$
$$\cancel{-40I_1 = 25}$$
$$\cancel{I_1 = 5/8\ A}$$

In FEAB

$$4I_1 - 2(I - I_1) - 2(I - I_1 - I_2) = 0$$
$$4I_1 - 2I + 2I_1 - 2I + 2I_1 + 2I_2 = 0$$
$$8I_1 - 4I + 2I_2 = 0$$
$$8I_1 + 4I_1 - 10 + 2I_2 = 0$$
$$-6I_1 + 5 = I_2$$

$$4I = 10 - 5/2 = 15/2$$
$$I = 15/8\ A$$

$$I_2 = -\frac{15}{4} + \frac{20}{4} = 5/4\ A$$

$$I_{AB} = \frac{15}{8} - \frac{5}{8} = \frac{10}{8}$$

$5 + 32 I_1 = 50$

$32 I_1 = 45$

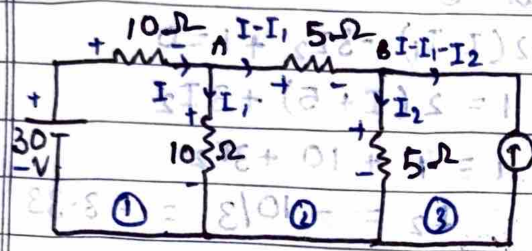
$I_1 = \frac{45}{32} = 1.40 A$

$I_2 = -3.4 A$

$I = 1.2 A$

$I_{AB} = I - I_1 - I_2 = 1.2 - 1.4 + 3.4 = 3.2 A$

Q



5 Ω resistor

In ①

$30 - 10I - 10I_1 = 0$

$I_1 + I = 3$ — ①

$I_1 = 3 - I$

In ②

$10 I_1 - 5I + 5I_1 - 5I_2 = 0$

$15 I_1 - 5I - 5I_2 = 0$

$45 - 15 I - 5I = 5I_2$

$9 - 4I = I_2$ — ②

In ③ since we have a current source so overall current in loop ③ is $-12 A$

$I - I_1 - I_2 = -12$

$I + I - 3 + 4I - 9 = -12$

$6I - 12 = -12$

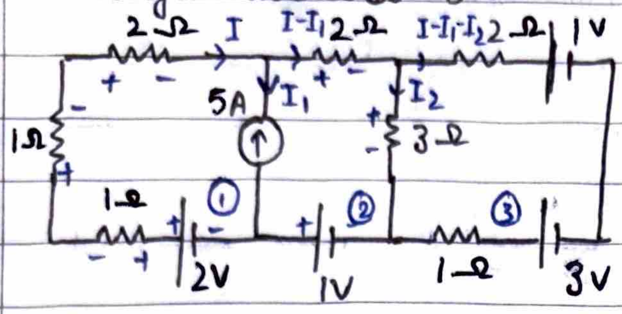
$I = 0 A$

$\Rightarrow I_2 = 9 A$

$\Rightarrow I_1 = 3 A$

$I_{5\Omega} = I - I_1 = -3A$

? Q voltage across 3 Ω



clearly $I_1 = -5A$

in loop 1

$-I - 2I + 2 - I = 0$

$2 = 4I$

$I = 0.5A$

in 2

$-2(I - I_1) - 3I_2 + 1 = 0$

$1 = 2(I + 5) + 3I_2$

$1 = 1 + 10 + 3I_2$

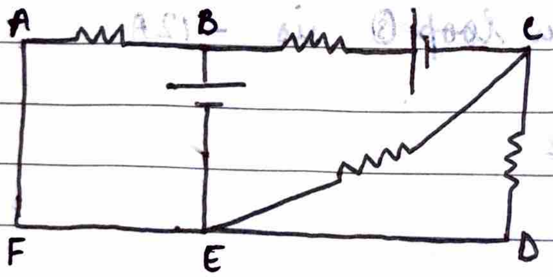
$I_2 = -10/3 = -3.33$

$V_{3\Omega} = I_{3\Omega} \times 3$

$= \frac{10}{3} \times 3 = 10V$

→ Junction & Node

Junction is where 3 or more than 3 branches combine but 2 or more than 2 combine at node

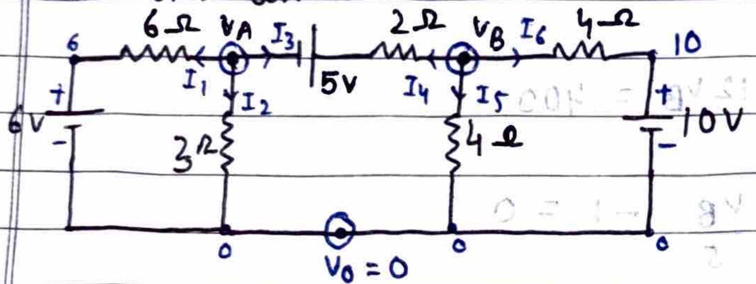


B → node
C → node
D → node
B & C → principle nodes

→ Steps to solve using nodal analysis

1. Identify principle nodes / junctions
2. Assign a junction potential wrt assigned reference junction value $V_0 = 0\,V$
3. Assuming all currents in outgoing direction from each junction from KCL eq
4. Solve to calculate junction potentials
5. Find required quantity.

Q Current in $3\,\Omega$



$I_1 = \dfrac{V_A - V_0 - 6}{6}$ ∵ $\dfrac{\text{Pot. diff}}{\text{Resistance}} = \dfrac{V_A - 6}{6}$

KCL at A

$$\frac{V_A - 6}{6} + \frac{V_A + 5 - V_B}{2} + \frac{V_A}{3} = 0$$

$$V_A - 6 + 3V_A + 15 - 3V_B + 2V_A = 0$$

$$6V_A - 3V_B + 9 = 0$$

$$2V_A + 3 = V_B$$

KCL at B

$$\frac{V_B}{4} + \frac{V_B - 10}{4} + \frac{V_B - 5 - V_A}{2} = 0$$

$$2V_B - 10 + 2V_B - 2V_A - 10 = 0$$

$$4V_B - 2V_A = 20$$

$$2V_B - V_A = 10$$

$$4V_A + 6 - V_A = 10$$

$$3V_A = 4$$

$$V_A = 4/3 \quad ⇒ V_B = 17/3\,V$$

$$I_{6\Omega} = \frac{V_A}{3} = \frac{4}{9}\,A$$

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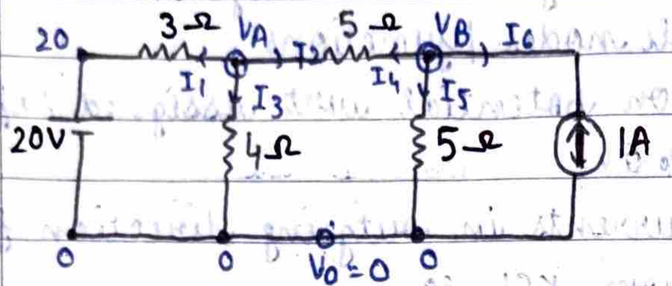
Q Current in 4 Ω branch



KCL at A

$$\frac{VA-20}{3} + \frac{VA}{4} + \frac{VA-VB}{5} = 0$$

$$\frac{20VA - 400 + 15VA + 12VA - 12VB}{60} = 0$$

$$47\,VA - 12\,VB = 400$$

$$\frac{-VA+VB}{5} + \frac{VB}{5} - 1 = 0$$

$$\cancel{\frac{VA}{5} = 1 \qquad VA = 5\,V}$$

$$\cancel{\Rightarrow \quad 235 - 12VB = 400}$$

$$\cancel{VB = -13.75\,V}$$

$$\cancel{I_{4\Omega} = \frac{VA}{4} = \frac{5}{4} = 1.25\,A}$$

$$2VB - VA - 5 = 0$$

On solving

$VA = 9.02\,V$

$VB = 2.012\,V$

$$I_{4\Omega} = \frac{VA}{4} = \frac{9.02}{4} = 2.25\,A$$

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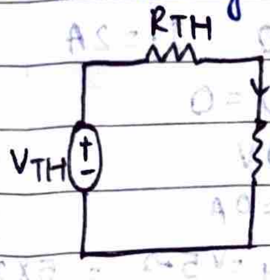
→ Thevenin's Theorem

Any linear bilateral network irrespective of it's complexities can be reduced into thevenin equivalent circuit having the Thevenin's open ckt voltage V_{TH} in series with Thevenin's equivalent resistance R_{TH} along with load resistance R_L



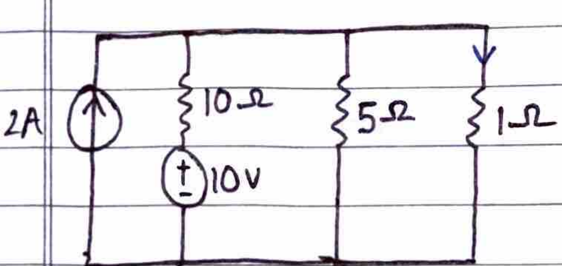
$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

→ Steps to solve

1. Identify load R_L (Jiska data nikalna h)
2. Remove R_L & calculate open ckt potential across two open ends. This will be V_{TH}
3. Again remove load resistance & replace all active sources by internal resistances
4. Calculate eq. resistance across open ends, its R_{TH}
5. Draw Thevenin eq ckt
6. Calculate I_L

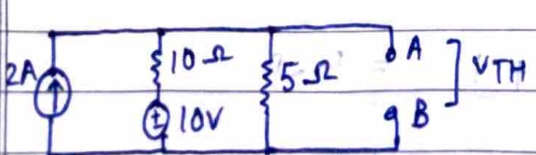
3. → voltage → close line
current → open ckt

Q Current via 1-Ω resistor



$R_L = 1\,\Omega$

Removing R_L



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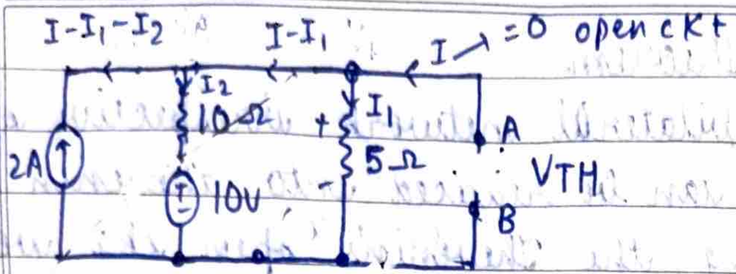


clearly $I - I_1 - I_2 = -2$ $\quad I = 0$ KCL $\Rightarrow I_1 + I_2 = 2$

$5I_1 - 10I_2 - 10 = 0$ $\quad I_1 = 2A$

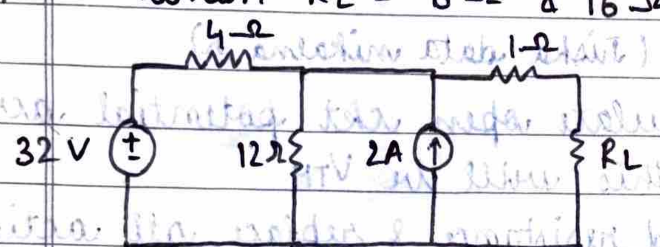
$5I_1 - 10I_2 = 10$

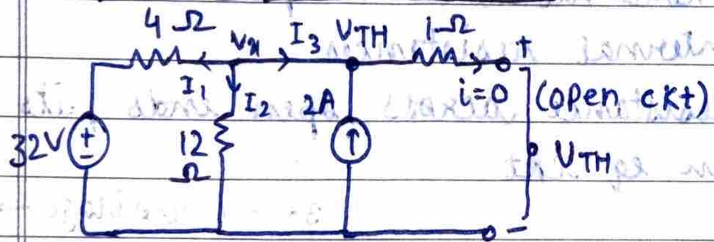
$I_1 - 2I_2 = 2$

$2 - I_2 - 2I_2 = 2$ $\quad I_2 = 0A$

$\Rightarrow V_{TH} = V_{5\Omega} = 5 \times 2 = 10V$

Q $\quad I$ when $R_L = 6\Omega$ & 16Ω





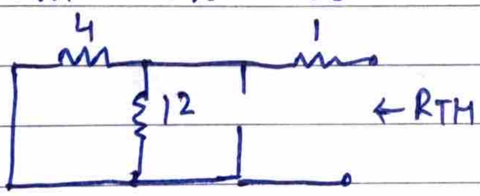
$$\frac{V_x - 32}{4} + \frac{V_x}{12} - 2 = 0$$

$$3V_x - 96 + V_x = 24$$

$$4x = 120$$

$$V_x = 30V$$

$$V_{TH} = V_x = 30\ V$$



$$R_{TH} = \frac{4 \times 12}{4 + 12} + 1 = 4\Omega$$

$$I_{6\Omega} = \frac{30}{10} = 3A \qquad I_{16\Omega} = \frac{30}{20} = 1.5\ A$$

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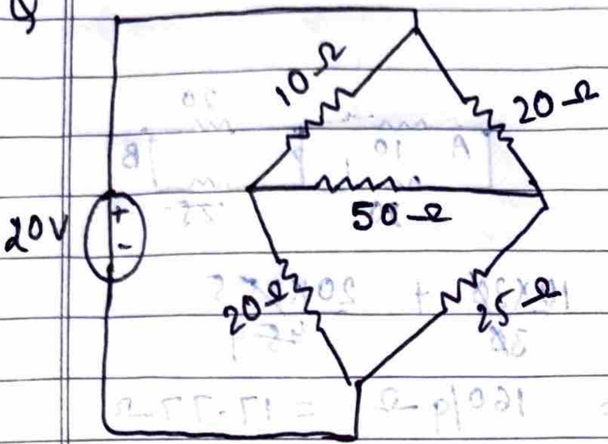
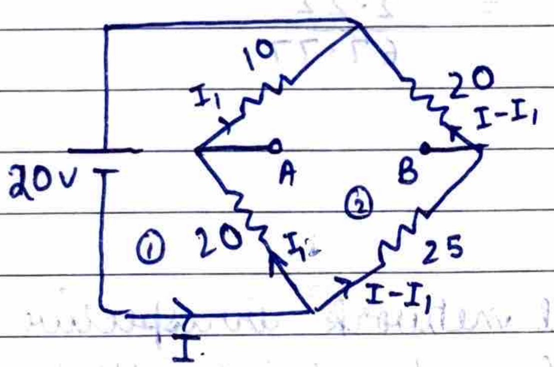
Q



$I_{50\Omega} = ?$

$R_L = 50\ \Omega$

Remove R_L



KVL in ①

$20 - 10I_1 - 20I_1 = 0$

$30I_1 = 20$

$I_1 = 2/3\ A$

$I_1 = 0.66A$

KVL in ②

$-20I_1 - 10I_1 + 20(I - I_1) + 25(I - I_1) = 0$

$-30I_1 + 20I - 20I_1 + 25I - 25I_1 = 0$

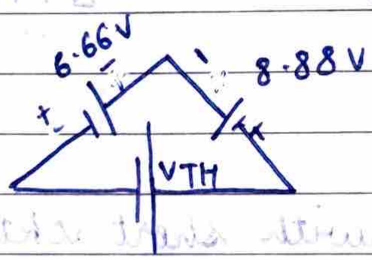
$-50I_1 - 25I_1 + 45I = 0$

$45I = \frac{75 \times 2}{3}$

$I = 10/9\ A$

$I = 1.11\ A$

$I - I_1 = 0.44$



$-V_{TH} - 6.66 + 8.88 = 0$

$-V_{TH} = -2.22\ V$

$V_{TH} = 2.22\ V$

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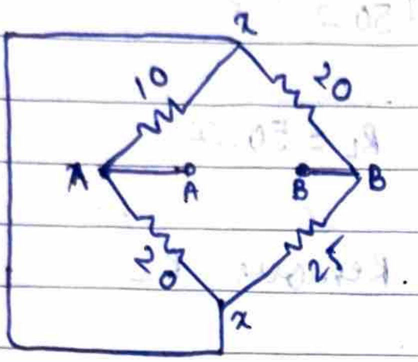
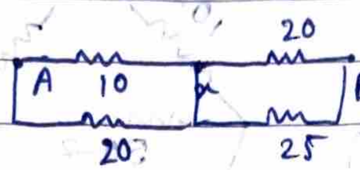




$$\frac{10\times 2\cancel{0}}{3\cancel{0}} + \frac{20\times \cancel{25}\,5}{\cancel{45}\,9}$$

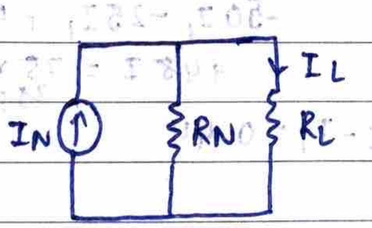
$$20/3 + 100/9 = 160/9\,\Omega = 17.77\,\Omega$$

$$\exists \quad I_L = \frac{2.22}{17.77+50} = \frac{2.22}{67.77}$$

$$= 0.032A$$

→ **Norton's Theorem**

Any linear bilateral network irrespective of it's complexicities can be reduced into Norton's equivalent circuit having Norton's short ckt current I_N in parallel with norton's equivalent resistance in parallel with R_L

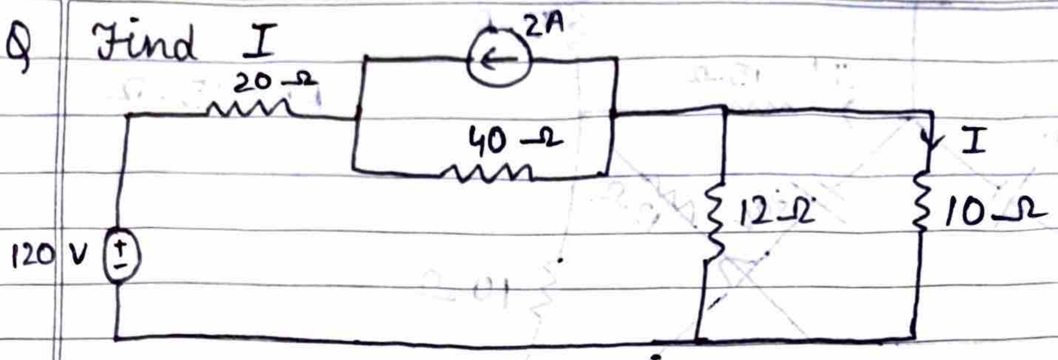


$$I_L = \frac{I_N R_N}{R_N + R_L}$$

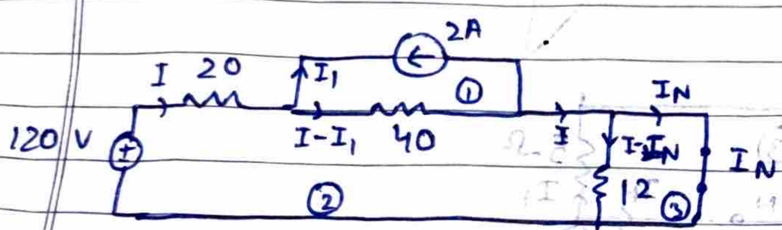
→ Steps

- Find R_L & replace it with short ckt branch
- current via this short ckted branch is I_N
- Remove R_L & replace the active sources by internal resistance & find R_N
- Draw Norton equivalent ckt
- calculate $I_L = \dfrac{I_N R_N}{R_N + R_L}$

Q Find I



$R_L = 10\,\Omega$



$I_1 = -2A$

$120 - 20I - 40(I - I_1) - 12(I - I_N) = 0$

$120 - 20I - 40I + 40I_1 - 12I + 12I_N = 0$

$120 - 72I - 80 + 12I_N = 0$

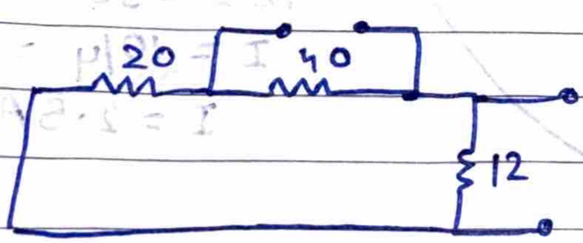
$40 = 72I - 12I_N$

$10 = 18I - 3I_N$ — ①

$12(I - I_N) = 0$

$I = I_N$

$\Rightarrow I = 2/3\ A = I_N$

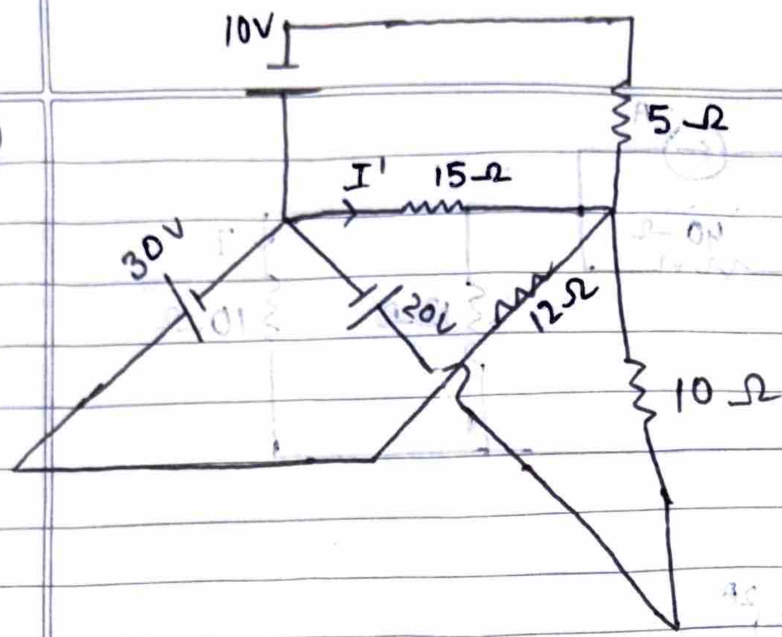


$R_N = \frac{60 \times \cancel{12}}{\cancel{72}\,6} = 10\,\Omega$

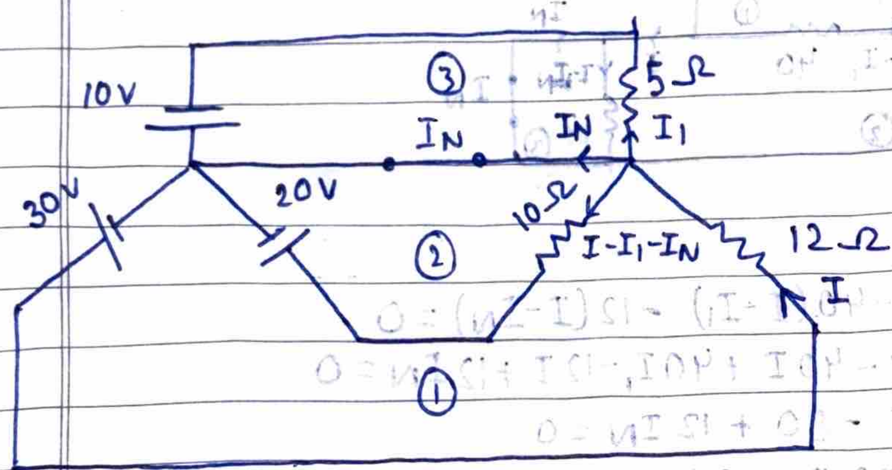
$I_L = \frac{2/3 \times 10}{10 + 10} = \frac{2/3 \times 1\cancel{0}}{2\cancel{0}}$

$= \frac{1}{3}\ A = 0.33\ A$

Q



$R_L = 15\,\Omega$



In ①

$-12I - 10(I - I_1 - I_N) - 20 + 30 = 0$

$-22I + 10I_1 + 10I_N + 10 = 0$

$-22I + 10I + 30 = 0$

$12I = 30$

$I = 10/4 = 5/2$

$I = 2.5A$

In ②

$20 + 10(I - I_1 - I_N) = 0$

$I_1 + I_N - I = 2$

$I_1 + I_N = I + 2$

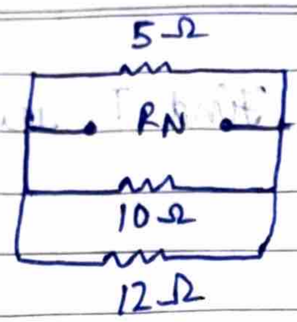
$\Rightarrow I_1 + I_N = 4.5A$

In ③

$-10 + 5I_1 = 0$

$I_1 = 2A$

$\Rightarrow I_N = 2.5A$



$5 \parallel 10 \rightarrow 10/3$ $\qquad$ $10/3 \parallel 12$

$$\frac{\frac{10}{\cancel{3}} \times 12}{\frac{10}{\cancel{3}} + \frac{36}{\cancel{3}}} = \frac{120}{46} = \frac{60}{23} = 2.608\,\Omega$$

$R_N = 2.608\,\Omega$

$$I_L = \frac{2.5 \times 2.608}{2.608 + 15} = \frac{2.5 \times 2.608}{17.608} = 0.370\,A$$

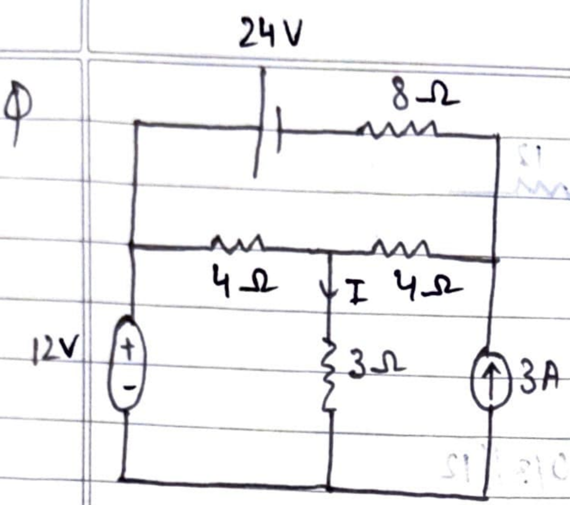
→ Super Position Theorem

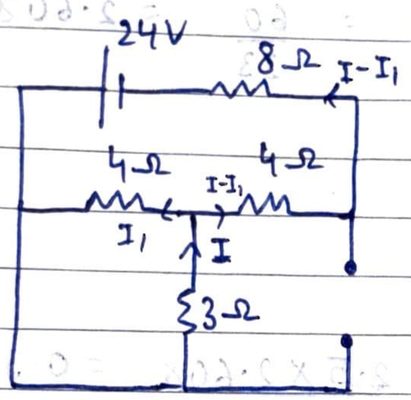
In any linear bilateral multisource network the current / voltage across any branch can be calculated by sum of values obtd. taking one source at a time & replacing other active sources by internal resistance

→ Steps

- Find branch, quantity & active source
- Take 1 & replace other by int resistance
- Calculate quantity for that source
- Repeat for all sources
- Sum algeraically

Q) Find I using SPT





$-24 + 8(I-I_1) + 4(I-I_1) - 4I_1 = 0$

$-24 + 12I - 16I_1 = 0$

$6 = 3I - 4I_1$

$-8I_1 = 6$

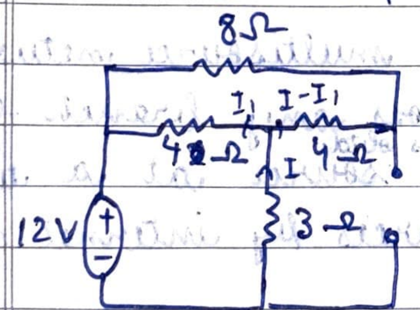
$I_1 = -3/4\ A$

$\underline{I = 1\ A}$

$-3I - 4I_1 = 0$

$3I + 4I_1 = 0$

$3I = -4I_1$



$-4(I-I_1) - 8(I-I_1) + 4I_1 = 0$

$3I - 3I_1 = I_1$

$4I_1 = 3I$

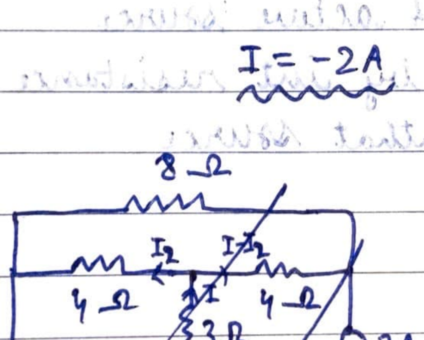
$-3I - 4I_1 - 12 = 0$

$3I + 4I_1 + 12 = 0$

$8I_1 = -12$

$I_1 = -3/2$

$\underline{I = -2A}$



$I_1 = +3A$

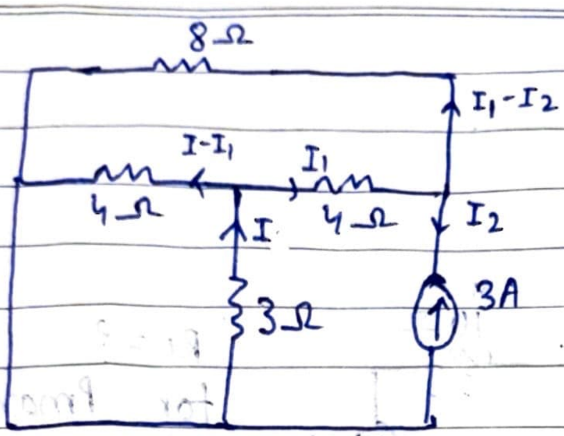
~~$-4I_2 + 8(I-I_2) + 4(I-I_2) = 0$~~

~~$I_2 = 3(I-I_2)$~~

~~$4I_2 = 3I$~~

$-3I - 4I_2 = 0$

$3I = -4I_2$



$I_2 = -3A$

$-4I_1 - 8(I_1 - I_2) + 4(I - I_1) = 0$

$-16I_1 + 8I_2 + 4I = 0$

$I + 2I_2 = 4I_1$

$I - 6 = 4I_1$

$I - 4I_1 = 6$ — (1)

$-6I = 6$

$I = -1A$

$3I + 4(I - I_1) = 0$

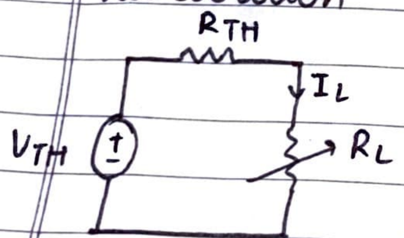
$7I = 4I_1$

By SPT $\quad I = 1 - 1 - 2 = -2A$

→ **Maximum Power Transfer Theorem**

The condition for maximum power to flow via load resistor R_L can be achieved when the R_L = Thevenin equivalent resistance of the ckt.

• **Derivation**



Power via R_L

$P = I_L^2 R_L$

By Thevenin's theorem

$I_L = \dfrac{V_{TH}}{R_{TH} + R_L}$

$$P = \left(\frac{V_{TH}}{R_{TH} + R_L}\right)^2 R_L$$

Diff wrt R_L & $= 0$ for P_{max}

$$\frac{dP}{dR_L} = V_{TH}^2\left[\frac{(R_{TH}+R_L)^2 - 2R_L(R_{TH}+R_L)}{(R_{TH}+R_L)^4}\right] = 0$$

$R_{TH} + R_L - 2R_L = 0$

$R_{TH} = R_L$ → This is req condition.

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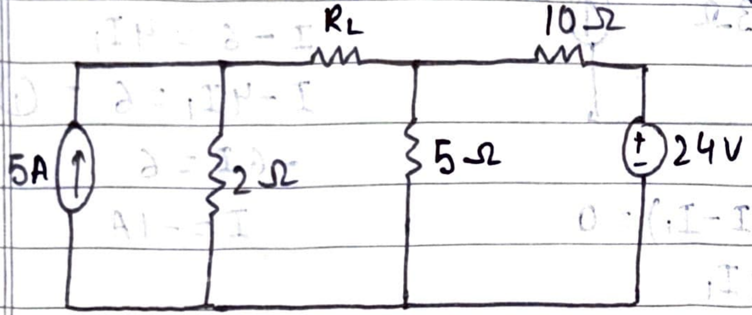
$\Rightarrow$ Power $= \dfrac{I_L^2 R_{TH}}{}$

$= \dfrac{V_{TH}^2}{4R_{TH}^2} R_{TH}$

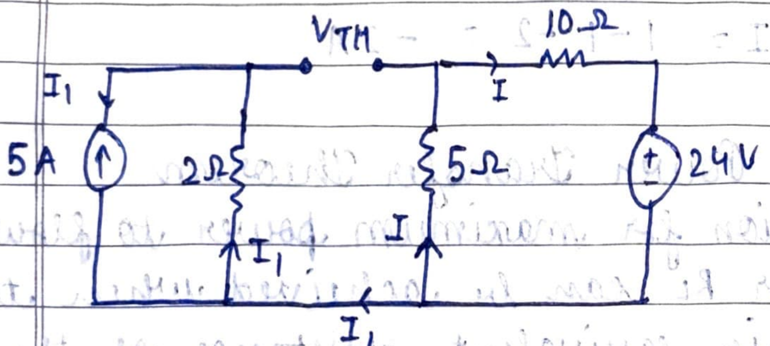
$P_{max} = \dfrac{V_{TH}^2}{4R_{TH}}$

Q



$R_L = ?$ for P_{max}

$P_{max} = ?$

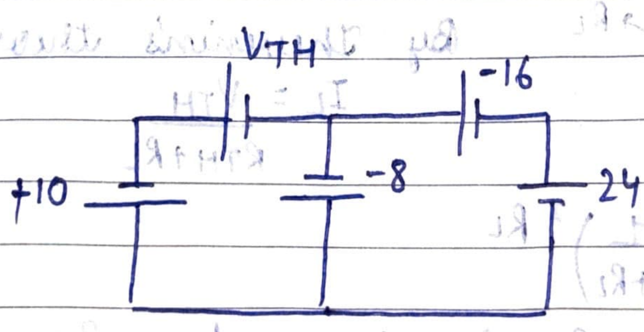


$I_1 = +5A$

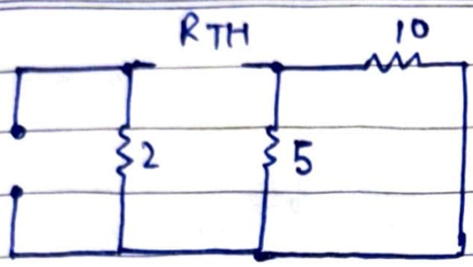
$-5I - 10I - 24 = 0$

$15I + 24 = 0$

$I = -8/5 = -1.6A$



$V_{TH} = 2V$



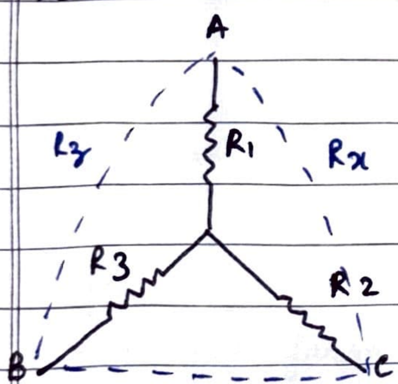
$$5 \| 10 + 2$$

$$\frac{10}{3} + 2 = \frac{16}{3}\ \Omega$$

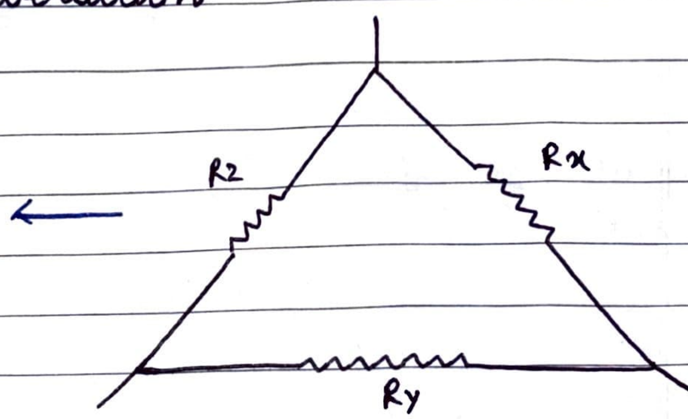
$$R_{TH} = 5.33\ \Omega$$

$$P_m = \frac{\cancel{(2)^2} \times 3}{\cancel{4} \times 16} = \frac{3}{16}\ W$$

→ Star - delta conversation



Star connection



Delta Transformation

$$R_x = R_1 + R_2 + \frac{R_1 R_2}{R_3}$$

$$R_y = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

$$R_z = R_1 + R_3 + \frac{R_1 R_3}{R_2}$$

For star to delta conversion